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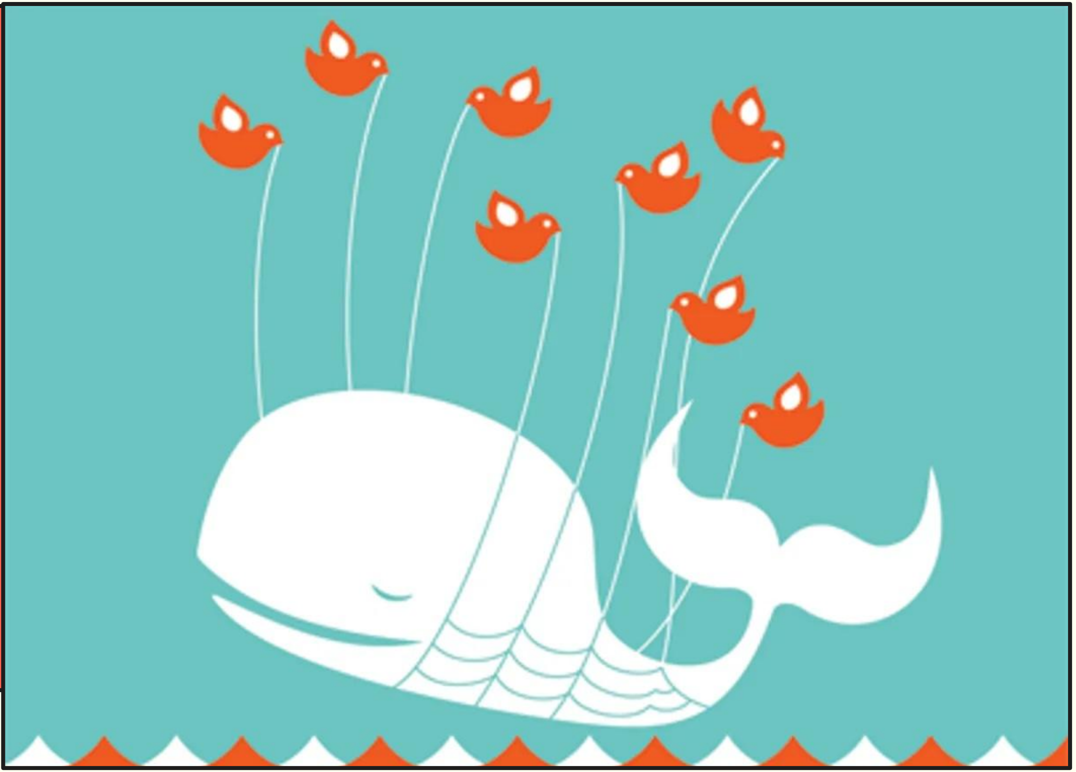
HUFS / Yonsei

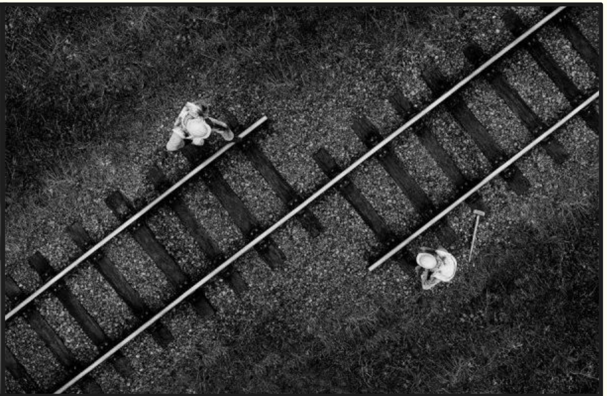
Validating Azure Pronunciation Assessment

Live Speech Test Demo

- Choose any text
- I'll read aloud and test my speech
- Demo Link Here!
- Later we can check the analysis PDF
- You can also test your own speech



Presentation Overview



The Problem & Potential Solutions

Neural network speech assessment is becoming more prevalent, but validation of these tools remains incomplete.

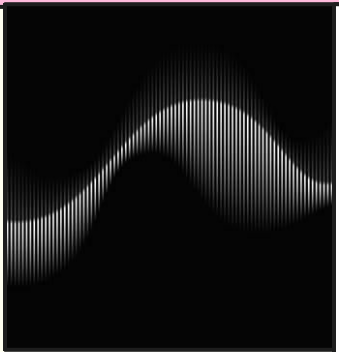
Methods & Measurement

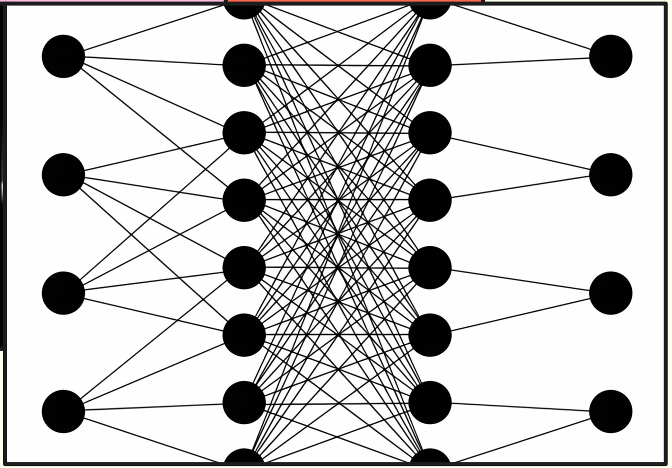
ICNALE corpus (n = 3,510)
& classroom data (n = 66)
Comparisons of Azure scores with linguistic indices and human ratings

Results & Future Directions

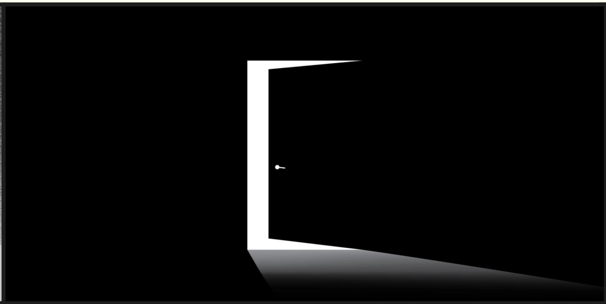
Strong rank-order alignment but large calibration differences

The Problem & Potential Solutions





Speech Assessment Challenges



Interrater variability

- Speech evaluation is inherently subjective
- Trained raters differ in applying scoring criteria
- Performance standards are complex and difficult to operationalize consistently

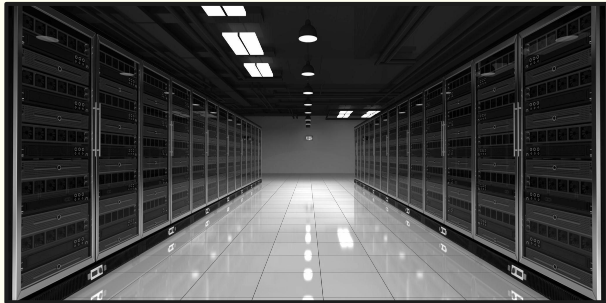
Limited independent validation

- Largely based on human–machine correlations
- Evidence often produced by test developers
- Rater characteristics and procedures are underreported

Accessibility constraints

- High cost and dependence on trained raters
- Limited interpretability of scores for teachers and learners

Speech Assessment Solutions



Commercial automated systems

- Widely deployed (e.g., SpeechRater, Versant, DET)
- Evolved from feature-based models to neural systems
- Scores derived from statistical mappings to human ratings

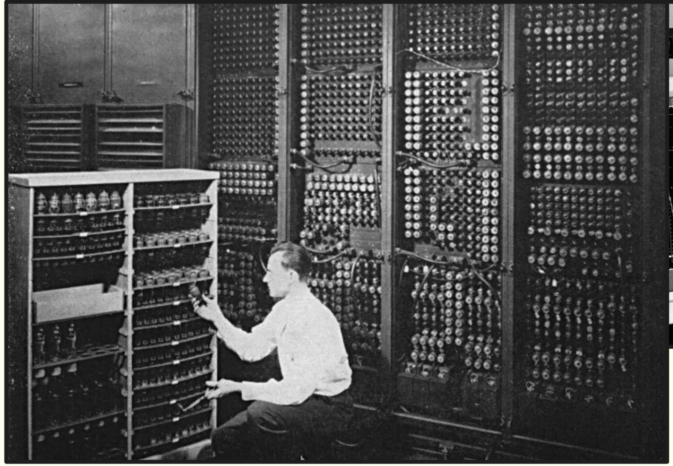
Experimental automated systems

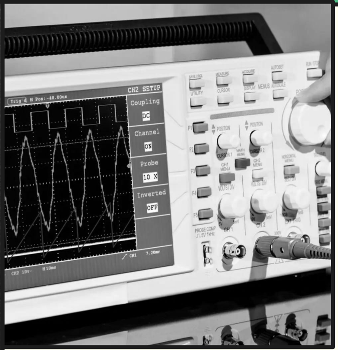
- Machine learning models approximate human speaking judgments
- Use of hierarchical and transformer-based architectures
- Strong predictive performance within datasets

Azure pronunciation assessment

- Production-level system for Accuracy, Fluency, Prosody
- Combines speech recognition features with multi-level modeling of speech patterns
- Evaluated primarily through correlations with expert ratings

Methods & Measurement





The Datasets

International Corpus Network of Asian Learners of English (ICNALE)

3,510 spoken monologues from learners across 10 countries, with CEFR levels (A2–B2+) assigned using external standardized benchmarks (e.g., TOEFL, TOEIC, IELTS).

Classroom contexts:
University EFL settings in China, Korea, and Japan using a controlled read-aloud task (Please Call Stella) under typical instructional conditions.

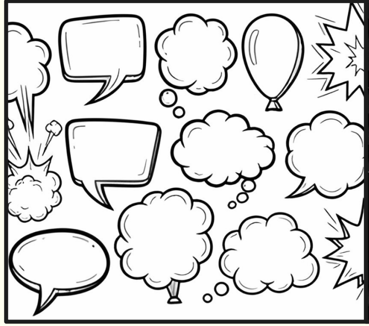
Classroom participants:
66 university-level EFL learners from China (28), Korea (20), and Japan (18).

Classroom raters:
Three experienced EFL professionals assigned scores on a calibrated 0–100 scale following a structured calibration procedure.

Linguistic Indices

TAALED

Tool for the Automatic Analysis of Lexical Diversity: Provides indices of lexical diversity reflecting the range and distribution of vocabulary use (e.g., MTLD, HD-D).

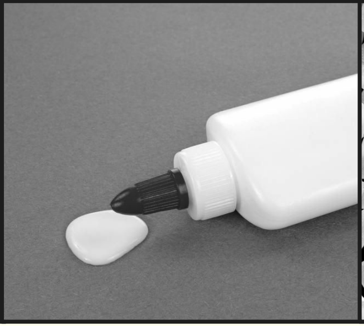


TAALES

Tool for the Automatic Analysis of Lexical Sophistication: Captures multi-dimensional lexical sophistication, including frequency, range, and age of acquisition.

TAACO

Tool for the Automatic Analysis of Cohesion: Quantifies discourse cohesion by capturing how ideas are linked through lexical overlap and semantic similarity.



TAASSC

Tool for the Automatic Analysis of Syntactic Sophistication and Complexity: Provides fine-grained indices of syntactic sophistication and usage-based grammatical patterns.

L2SCA

L2 Syntactic Complexity Analyzer: Operationalizes syntactic complexity using clausal and phrasal structural measures.

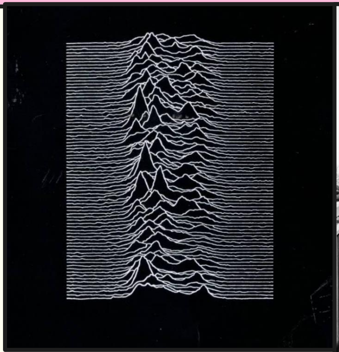
Prosody, Accuracy & Fluency

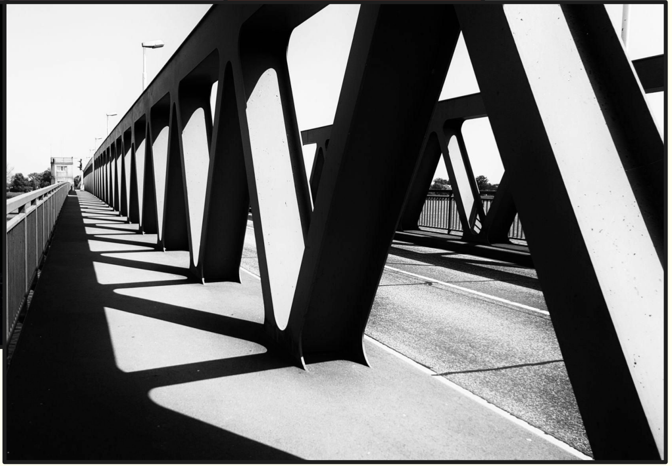
Prosody
- Stress, rhythm, and intonation
- Helps listeners process meaning

Accuracy
- Segmental control
  (vowels, consonants, clusters)
- Contrasts necessary for intelligibility

Fluency
- Temporal control (rate, pauses, repairs)
- Reflects processing efficiency

Results & Future Directions





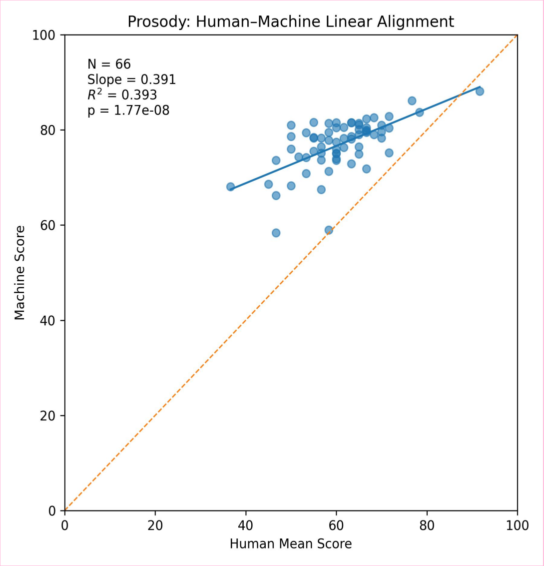
Correlates in Context

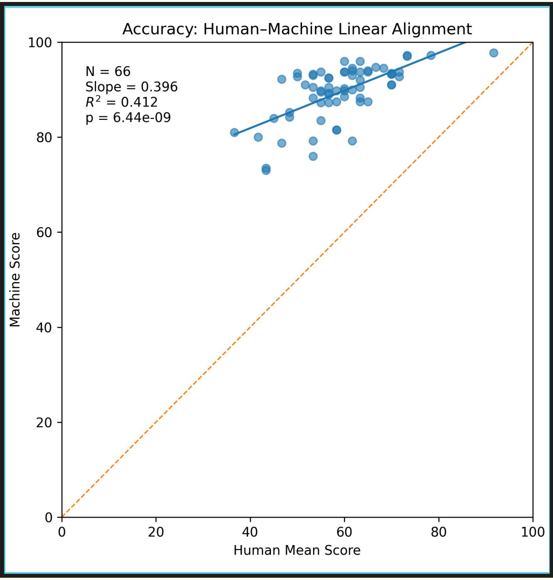
CEFR Correlates	Spearman's ρ
1. **TAALED** basic ntypes	**0.265**
2. **AZURE** prosody	**0.261**
3. **AZURE** overall pronunciation	**0.256**
4. **AZURE** accuracy	**0.242**
5. **TAALED** basic ntokens	**0.241**
6. **AZURE** completeness	**0.230**
7. **TAALED** MTLD	**0.213**
8. **TAALED** MATTR 50	**0.212**
9. **AZURE** fluency	**0.186**

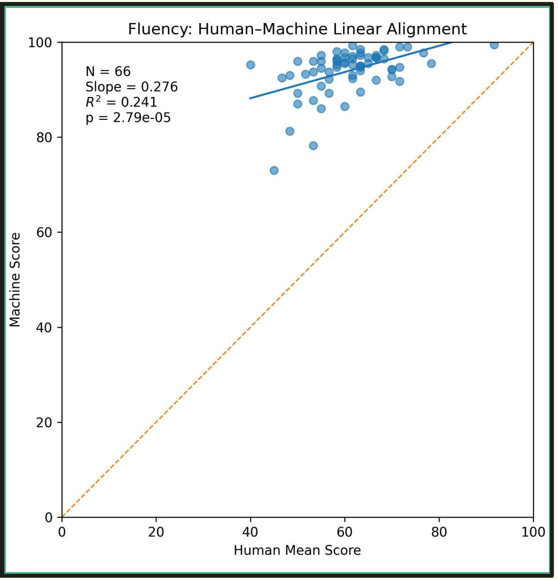
Note:

These are Spearman's ρ correlations (all $p < .001$) between CEFR proficiency and selected spoken-task indicators in the ICNALE corpus. Token and type counts are included as reference benchmarks, consistent with prior spoken lexical diversity research (e.g., Kyle et al.), in which MATTR and MTLD are evaluated relative to these foundational measures.

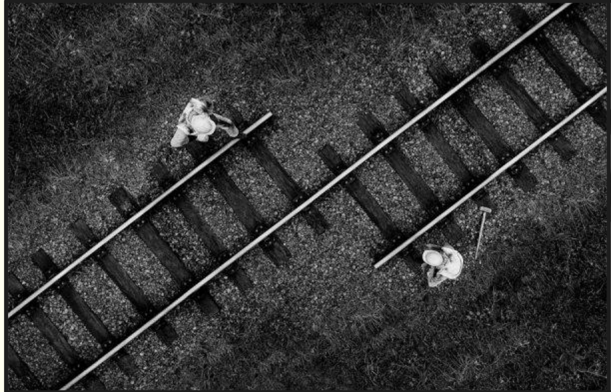
Construct Validation







Discussion



Signal, Scale, and Fluency

- **Strong rank-order agreement** (robust re: accuracy & prosody)
- **Systematic bias** (misalignment between human & machine scores)
- **Fluency measurement issues** (rater variability & misalignment)

Limitations & Future Research

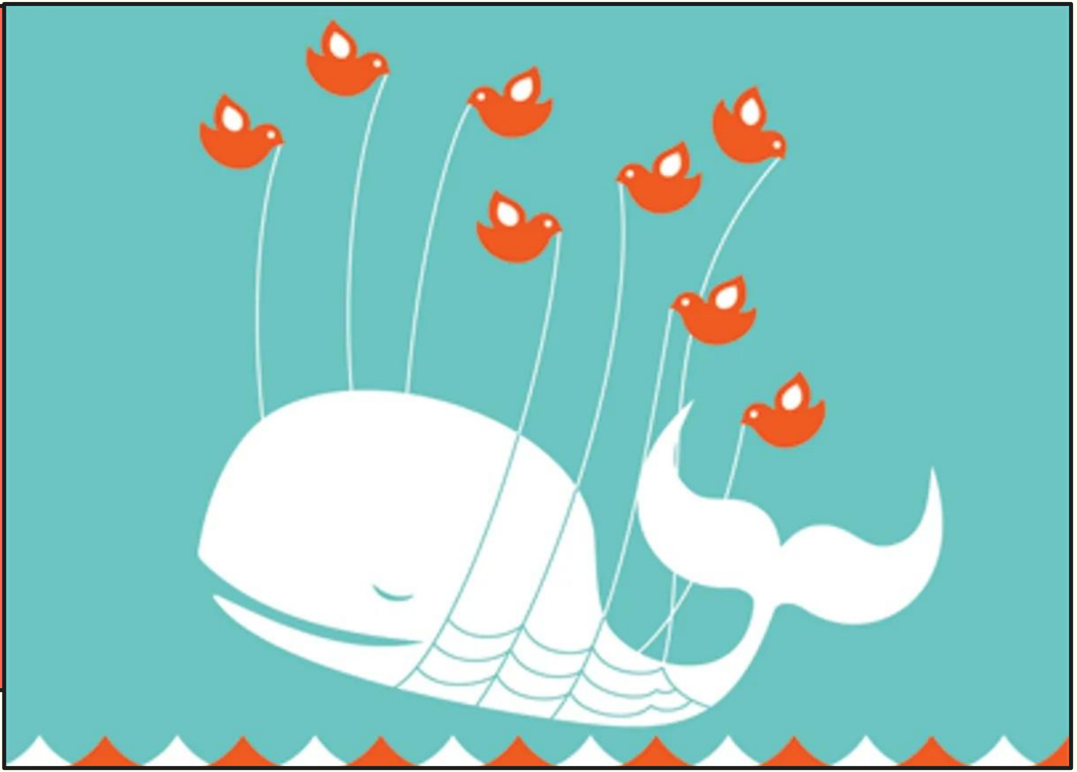
- **Monologue task only** (extend to diverse/interactive tasks)
- **Human rating variability** (refine constructs & link scores)

Implications for Practice

- **Useful feedback** for **formative use**
- **Supports ipsative tracking** (change over time)
- **$1/hr, 33 languages**
- **Requires calibration** or **human oversight** for **high-stakes use**

Live Speech Test Demo

- Let's check the results of the demo!
- Demo Link Here!
- Demo Results Here!
- Backup Results (ㅠㅠ)



Q&A

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